TECHNICAL DOCUMENTATION

A2A Travel Management Agent System

Powered by Google AI Agent-to-Agent Framework

8 AI Agents	End-to-End	Real-Time
Orchestrated	Automation	Expense Tracking

Version:	Audience:	Platform:	Classification:
1.0	Operations Team	Google AI / GCP	Confidential

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1. Executive Summary

This document provides a comprehensive technical specification for the A2A (Agent-to-Agent) Travel Management System, purpose-built to automate and streamline client visit logistics for organizational travel. The system leverages Google AI's multi-agent orchestration framework to deliver an intelligent, autonomous, and fully trackable travel management experience.

The platform coordinates eight specialized AI agents that communicate in real-time to handle every aspect of a business trip — from initial travel request through flight and hotel booking, local transport arrangement, on-ground guide assistance, emergency support, and granular expense categorization and reporting.

Key Business Value

The A2A Travel Management System eliminates manual coordination overhead, enforces organizational travel policies automatically, provides end-to-end expense visibility, and ensures traveler safety through proactive monitoring and emergency response capabilities.

1.1 Objectives

- Fully automate end-to-end travel booking and coordination for client visits
- Enforce corporate travel policies and budget thresholds in real-time
- Provide 360-degree expense visibility with automatic personal/business categorization
- Ensure traveler safety through emergency contact management and proactive alerts
- Deliver actionable post-trip analytics for finance and operations teams
- Reduce travel management overhead by 70% through AI-driven automation

1.2 Scope

This system covers all domestic and international client visit travel initiated by the organization. It handles pre-trip planning, in-transit support, on-ground assistance, and post-trip reconciliation. Personal expenses are automatically excluded from corporate reporting with full audit trails maintained throughout.

2. System Architecture

2.1 Google A2A Framework Overview

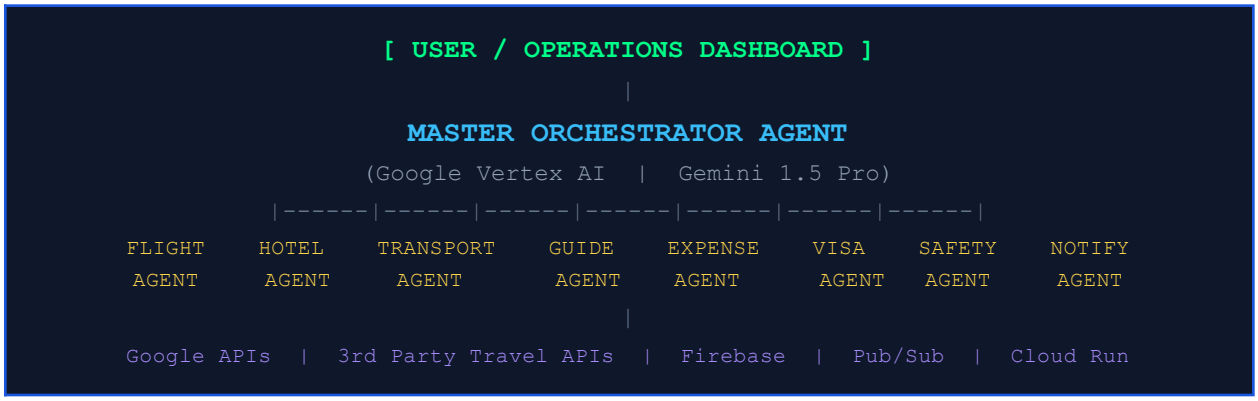
The system is built on Google AI's Agent-to-Agent (A2A) protocol, which enables multiple specialized AI agents to collaborate autonomously. Each agent exposes an Agent Card (a structured manifest) and communicates via standardized JSON-RPC 2.0 messages over HTTPS. The Orchestrator Agent acts as the central coordinator, delegating tasks to specialist agents and aggregating responses into cohesive user experiences.

Google A2A Protocol

The A2A protocol standardizes how AI agents discover each other (via /.well-known/agent.json endpoints), negotiate capabilities, delegate tasks using structured Task objects, stream progress via Server-Sent Events (SSE), and exchange results in a provider-agnostic format. This ensures all eight agents in this system can interoperate regardless of their underlying model or hosting environment.

2.2 High-Level Architecture

The architecture follows a hub-and-spoke model with the Master Orchestrator at the center. All user interactions enter through the Orchestrator, which routes tasks to appropriate specialist agents, collects responses, and synthesizes unified outputs back to the user and operations dashboard.



2.3 Infrastructure Components

Component	Google Service	Purpose
Agent Hosting	Cloud Run (Serverless)	Each agent runs as an isolated, scalable container
Agent Registry	Vertex AI Agent Builder	Central agent discovery and routing

Component	Google Service	Purpose
LLM Backbone	Gemini 1.5 Pro / Flash	Natural language understanding and reasoning
Database	Firestore + BigQuery	Real-time trip data and analytics warehouse
Messaging	Cloud Pub/Sub	Async agent-to-agent task delegation
File Storage	Cloud Storage (GCS)	Receipts, documents, itineraries
Secrets	Secret Manager	API keys, credentials vault
Observability	Cloud Logging + Trace	Full agent audit trail
Auth	Identity-Aware Proxy (IAP)	Zero-trust access to agent endpoints
Notifications	Firebase Cloud Messaging	Push alerts to traveler mobile app

3. Agent Catalog

Each agent in the system is an independent AI microservice with its own Agent Card, capabilities manifest, and API endpoints. Agents communicate via the A2A protocol and can be independently scaled, updated, or replaced without impacting the rest of the system.

Icon	Agent Name	Primary Responsibility
✈	**Flight Booking Agent**	Searches, compares, and books flights. Handles cancellations, rescheduling, seat preferences, and airline loyalty integration.
🏨	**Hotel Booking Agent**	Finds and reserves accommodations based on location, budget, ratings, and proximity to client site. Manages check-in/out changes.
🚗	**Local Transport Agent**	Arranges airport transfers, cab bookings, car rentals, and metro passes. Tracks all ground transport for expense reporting.
🗺	**Travel Guide Agent**	Provides destination knowledge: local emergency contacts, hospitals, police, embassy, cultural norms, weather, and area safety scores.
💰	**Expense Tracker Agent**	Captures and categorizes every expense in real-time. Auto-separates personal vs. business spend. Generates policy-compliant reports.
📋	**Visa & Docs Agent**	Checks visa requirements, tracks application deadlines, validates document completeness, and stores travel documents securely.
🔔	**Safety & Alerts Agent**	Monitors flight delays, political advisories, weather disruptions, and health alerts. Triggers proactive notifications and escalations.
📢	**Notification Agent**	Unified communication layer. Sends itinerary updates, reminders, approvals, and emergency alerts via Email, SMS, WhatsApp, and push.

3.1 Master Orchestrator Agent

The Orchestrator is the entry point for all travel requests. It parses natural language inputs, determines which specialist agents to invoke, sequences multi-agent workflows, and consolidates responses into coherent outputs for the user and the operations dashboard.

Agent Card Specification

Agent ID	travel-orchestrator-v1
Endpoint	https://agents.company.com/orchestrator
Model	gemini-1.5-pro-002

Input Modalities	Text, Voice (via Google STT), File Upload
Skills	travel_planning, trip_coordination, policy_enforcement, multi_agent_routing
Auth	OAuth 2.0 + IAP
Max Concurrent Tasks	500 per instance | Auto-scales via Cloud Run

Orchestration Workflow

1. Travel Request Intake: User submits trip request via app, web portal, or natural language prompt
2. Intent Parsing: Gemini extracts trip parameters (destination, dates, traveler, client, budget)
3. Policy Validation: Trip checked against travel policy rules before any bookings
4. Agent Delegation: Parallel tasks sent to Flight, Hotel, Transport, Guide, and Visa agents
5. Approval Routing: High-value bookings routed to approver via Notification Agent
6. Confirmation & Itinerary: Consolidated itinerary generated and delivered to traveler
7. Live Monitoring: Safety Agent monitors trip in real-time; alerts triggered on anomalies
8. Post-Trip: Expense Agent compiles final report; Firestore and BigQuery updated for analytics

3.2 Flight Booking Agent

The Flight Booking Agent integrates with global distribution systems (GDS) and direct airline APIs to search, compare, and book flights in real-time. It applies corporate travel policies automatically and selects the most cost-effective compliant option unless the traveler specifies preferences.

Supported Integrations

- Amadeus GDS API — Global flight inventory search
- Sabre Travel API — Fare comparison and booking
- Google Flights API — Real-time price intelligence
- Direct airline APIs (IndiGo, Air India, Emirates, Singapore Airlines, etc.)
- Google Calendar — Automatic flight event creation
- PDF generation — E-ticket delivery via Cloud Storage + email

Key Capabilities

Capability	Implementation
Multi-city routing	Automatically builds optimal multi-leg itineraries
Fare class enforcement	Economy enforced unless policy exception approved

Capability	Implementation
Refund policy check	Only refundable/flexible fares booked for new trips
Layover optimization	Avoids connections < 60 min or > 4 hours by default
Price alert monitoring	Monitors fares post-booking; suggests rebooking if savings > 20%
Loyalty program integration	Applies corporate frequent flyer numbers automatically
Cancellation automation	Cancels and initiates refund on trip cancellation event

3.3 Hotel Booking Agent

The Hotel Booking Agent sources accommodations from OTA APIs and direct hotel chains, applying nightly rate caps, star-rating requirements, and proximity rules relative to the client visit site. It handles all modifications and late check-out requests autonomously.

Supported Integrations

- Booking.com API — Global hotel inventory
- Expedia Partner Solutions API — Rates and availability
- Google Hotels API — Location-based discovery and reviews
- MakeMyTrip / Agoda API — Regional hotel access
- Direct chain APIs (Marriott, Hilton, IHG, Taj, Hyatt)

Booking Decision Parameters

Parameter	Default Value	Notes
Max nightly rate (domestic)	INR 8,000	Configurable per city tier
Max nightly rate (international)	USD 200	Configurable per region
Minimum star rating	3 stars	4-star for VP level and above
Max distance from client site	5 km	Overridden if no options available
Breakfast included preference	Yes	Reduces per-diem claim complexity
Corporate rate priority	Yes	Negotiated rates applied first
Free cancellation required	Yes (within 24 hrs)	Hard policy for all bookings

3.4 Local Transport Agent

The Local Transport Agent manages all ground transportation throughout the trip lifecycle. It coordinates airport pickups, intercity travel, daily commutes to client sites, and return journeys, ensuring every movement is logged for expense tracking.

Transport Modalities Supported

- Airport transfers: Pre-booked cabs via Uber for Business / Ola Corporate APIs
- Car rental: Integration with Zoomcar, Hertz, Avis for multi-day rentals
- Metro and public transit: Google Maps Transit API for routes and fare estimates
- Intercity bus/train: IRCTC API (India), Trainline API (Europe) for rail bookings
- Executive chauffeur: Premium options for client-facing transport days

Expense Auto-Tagging

Every transport booking made through this agent is automatically tagged with trip ID, traveler ID, date, purpose (Airport Transfer / Client Site / Personal), and amount. Personal trips (evenings, weekends) are flagged separately based on time and destination pattern analysis.

3.5 Travel Guide Agent

The Travel Guide Agent serves as an intelligent on-ground companion for travelers. It proactively delivers destination-specific information and maintains an emergency response capability throughout the trip duration.

Information Delivered at Trip Start

Category	Information Provided
Emergency Contacts	Local police, ambulance, fire, nearest hospital, 24hr clinic
Embassy / Consulate	Nearest embassy address, phone, emergency hotline
Healthcare	Hospital names, ratings, addresses within 5km; travel insurance helpline
Local Laws	Dress code, photography rules, prohibited items, alcohol laws
Currency & Payments	Current exchange rate, ATM locations, card acceptance norms
Communication	Local SIM providers, roaming tips, WiFi hotspots at hotel/airport
Weather Brief	7-day forecast, seasonal clothing recommendations
Safety Score	Neighborhood safety ratings sourced from Numbeo and government data

Category	Information Provided
Cultural Etiquette	Business greeting norms, dining etiquette, meeting culture
Client Site Intel	Parking, entry requirements, dress code, point of contact

3.6 Expense Tracker Agent

The Expense Tracker Agent is the financial intelligence layer of the system. It operates continuously during the trip to capture, categorize, and validate every expense, ensuring policy compliance and eliminating the manual expense reporting burden from travelers.

Expense Capture Methods

- Receipt OCR: Traveler photographs receipt via mobile app; Gemini Vision extracts merchant, amount, date, category
- Automatic import: Booked expenses (flights, hotels, cabs) auto-imported from booking agents
- Card integration: Corporate card transactions sync in real-time via Visa/Mastercard APIs
- Per-diem auto-calculation: Daily allowances computed by destination and meal type
- Manual entry: Fallback for cash expenses with category selection

Expense Categories

Category	Sub-Category	Auto-Detection Logic
Air Travel	Domestic / International	Flight booking agent data; airline merchant codes
Accommodation	Hotel / Serviced Apt	Hotel booking agent data; lodging merchant category
Ground Transport	Cab / Rail / Car Rental	Transport agent data; MCC 4111, 7512, 4131
Meals	Breakfast/Lunch/Dinner/Snacks	Time-of-day + restaurant MCC codes
Client Entertainment	Meals / Events	Manually flagged or inferred from group spend
Communication	SIM / Data / Calls	Telecom merchant codes
Visa & Travel Docs	Fees / Attestation	Government / embassy merchant codes
PERSONAL	All personal spend	Flagged and excluded from corporate report

Personal vs. Business Separation Logic

The agent uses a multi-signal classifier to identify personal expenses:

- Location signal: Expenses at non-business locations (entertainment zones, shopping malls, personal dining on non-client days)
- Time signal: Weekend charges or charges after 9 PM on non-event days
- Category signal: Purchases at categories with zero business relevance (spa, personal shopping, movies)
- Traveler flag: Manual personal expense flag by traveler in mobile app
- Policy rule: Amount exceeds meal per-diem cap — excess automatically marked personal

Audit Guarantee

Every expense decision — whether categorized as business or personal — is logged with the classification reason, confidence score, and agent version. Finance teams can review, override, and annotate any classification. Full audit trail retained in BigQuery for 7 years.

3.7 Visa & Documentation Agent

This agent ensures all travel documents are in order before trip commencement, preventing last-minute compliance failures that could disrupt client visits.

- Visa requirement check: Queries Sherpa API and government databases for destination-specific visa rules
- Application deadline tracking: Calculates required processing time and sets escalating reminders
- Document checklist: Generates personalized checklist (passport validity, invitation letter, hotel proof, etc.)
- Secure storage: All uploaded documents stored in encrypted GCS with 90-day retention
- Travel insurance validation: Confirms policy covers destination, duration, and medical emergencies
- Passport expiry alert: Warns if passport expires within 6 months of travel date

3.8 Safety & Alerts Agent

The Safety Agent monitors 15+ data sources in real-time to protect travelers and enable proactive organizational response to disruptions.

Monitored Data Sources

Source	Alert Type Triggered
FlightAware / FlightRadar24	Flight delays, diversions, cancellations
NOAA / IMD Weather APIs	Severe weather at origin, transit, or destination

Source	Alert Type Triggered
US State Dept / MFA Advisories	Travel advisories, security threats
WHO / CDC Health Feeds	Disease outbreaks, vaccination requirements
Local news sentiment analysis	Civil unrest, strikes, public safety events
Airport status APIs	Immigration backlogs, terminal closures
Hotel security feeds	Property-specific incidents reported

Escalation Protocol

9. Level 1 (Info): Push notification to traveler app — flight delay < 2 hours, weather advisory
10. Level 2 (Action Required): SMS + email to traveler and travel desk — rebooking triggered automatically
11. Level 3 (Emergency): Calls to traveler, manager, and security desk — incident report opened in Firestore

4. Data Flows & Agent Communication

4.1 New Trip Request Flow

The following describes the complete data flow when a new travel request is submitted. All agent-to-agent communication uses the A2A Task object format with structured JSON payloads over HTTPS.

Step	Action	Agent(s) Involved	Output
1	User submits travel request with destination, dates, traveler name	Orchestrator	Parsed trip intent JSON
2	Policy check against budget, class, advance booking rules	Orchestrator	Policy validation result
3	Parallel: flight options fetched; hotel options fetched	Flight + Hotel Agents	Top 3 options each
4	Visa and document requirements checked	Visa Agent	Document checklist
5	Traveler guide brief prepared for destination	Guide Agent	Destination briefing PDF
6	Options presented for approval (auto or manual)	Orchestrator + Notify	Approval request sent
7	Bookings confirmed on approval	Flight + Hotel + Transport	Booking confirmations
8	Consolidated itinerary + guide package delivered	Orchestrator + Notify	Itinerary doc + push alert
9	Trip record created in Firestore; monitoring begins	Safety Agent	Monitoring active

4.2 A2A Message Format

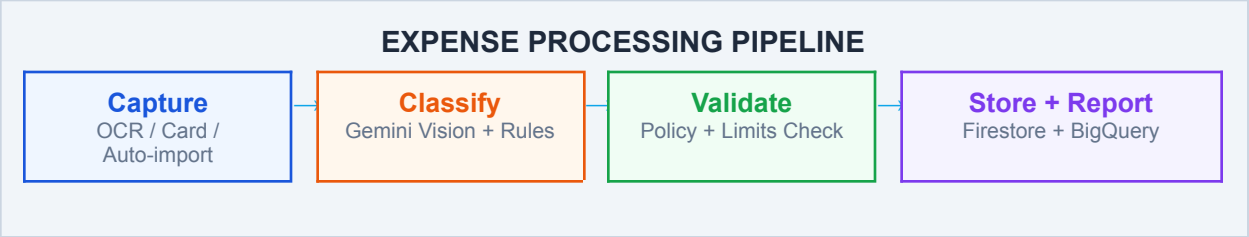
All inter-agent communication uses the following standardized Task object structure as defined by the Google A2A specification:

```
// A2A Task Object — Standard Format
{
  "id": "task-uuid-v4",
  "sessionId": "trip-2024-CLIENT-001",
```

```
  "status": { "state": "submitted" },
  "message": {
    "role": "user",
    "parts": [{ "type": "text", "text": "Book flight DEL-BOM 15-Jan 8AM,
economy" }]
  },
  "metadata": {
    "travelerId": "EMP-1042",
    "tripId": "TRIP-2024-0115-01",
    "clientId": "CLIENT-ACME-007",
    "budgetCode": "CC-SALES-2024"
  }
}
```

4.3 Expense Data Flow

The Expense Tracker Agent maintains a continuous data pipeline throughout the trip lifecycle. Every financial transaction — whether auto-imported from booking agents or captured via receipt OCR — enters a classification pipeline before being recorded.



5. Operations Team Guide

This section is written specifically for the travel operations team. It covers daily workflows, dashboard usage, exception handling, and approval management.

5.1 Operations Dashboard

The Operations Dashboard is a real-time web application built on Google Looker Studio, backed by BigQuery. It provides full visibility into all active and historical trips.

Dashboard Modules

Module	What You Can Do
Trip Command Center	View all active trips in real-time; see traveler location, current leg, and next action
Approval Queue	Review and approve pending bookings exceeding policy thresholds
Expense Dashboard	Live expense tracker per trip; drill down to individual receipts
Alert Center	All active safety alerts; one-click escalate to traveler or emergency contact
Document Vault	Search and retrieve all travel documents for any trip or traveler
Policy Manager	Update travel policy rules (rate caps, class limits, advance booking days)
Analytics & Reports	Post-trip spend reports, savings vs. market rate, policy compliance %

5.2 Approvals Workflow

The system supports two approval modes: fully automated (for in-policy bookings) and human-in-the-loop (for policy exceptions). Operations team manages all exception approvals.

Auto-Approval Rules

Bookings that meet ALL of the following are auto-approved: advance booking > 7 days, total cost within budget code limit, fare class = economy (domestic) or economy/business per policy (international), hotel star rating and rate within caps, traveler has active trip record in system.

Manual Approval Triggered When

Any of these conditions trigger manual review: booking cost exceeds trip budget by >15%, international business class request, hotel rate exceeds cap, trip duration > 14 days, destination under travel advisory Level 2+, same-day or last-minute booking, client entertainment spend > INR 5,000.

5.3 Exception Handling Procedures

Flight Delay / Cancellation

12. Safety Agent auto-detects delay and notifies traveler within 2 minutes of airline system update
13. If delay > 3 hours, Flight Agent auto-searches next available options and presents to traveler
14. Traveler selects alternative; rebooking executed automatically by Flight Agent
15. Hotel Agent extends hotel stay if overnight delay, notifies operations team
16. Operations team reviews rebooking cost delta; approves if > 20% above original fare

Traveler Emergency

17. Traveler presses SOS button in app OR Safety Agent detects distress signal
18. Orchestrator initiates Level 3 escalation: simultaneous call to traveler, manager, security desk
19. Guide Agent sends nearest hospital and embassy contacts to all parties
20. Operations team opens incident record; coordinates with local emergency services
21. Travel insurance contact triggered; all documentation auto-compiled for claim

5.4 Post-Trip Expense Reconciliation

Within 24 hours of trip return, the Expense Tracker Agent generates a draft expense report and sends it to the traveler for review. The operations team receives the finalized report for finance submission.

Day	Action	By Whom	System Action
T+0	Trip ends; return leg confirmed	Safety Agent auto-detects	Expense report draft generated
T+1	Traveler reviews draft; flags personal items	Traveler via mobile app	Personal items excluded from report
T+2	Traveler submits final report	Traveler	Report locks; sent to ops team
T+3	Ops reviews; approves or queries	Operations Team	Approved report sent to finance
T+5	Finance processes reimbursement	Finance Team	BigQuery updated; trip closed

6. Security & Compliance

6.1 Security Architecture

- Zero-Trust Network: All agent-to-agent communication requires mutual TLS and IAP verification
- Data Encryption: All data encrypted at rest (AES-256) and in transit (TLS 1.3)
- PII Handling: Traveler personal data tokenized in Firestore; raw data never exposed in logs
- Secret Management: All API keys and credentials stored in Google Secret Manager with rotation policies
- RBAC: Role-based access control enforced at dashboard, API, and agent levels
- Audit Logging: Every agent action, booking, approval, and expense decision logged immutably in Cloud Logging

6.2 Compliance Standards

Standard	Coverage
GDPR / DPDP Act (India)	Traveler consent captured; right to erasure implemented; data residency in-region
ISO 27001	Information security management; annual audit and penetration testing required
SOC 2 Type II	GCP infrastructure inherits SOC 2 controls; annual third-party audit
PCI-DSS	Corporate card data never stored in system; tokenized via payment gateway
Internal Travel Policy	All bookings auto-validated against corporate travel policy v2.4 before execution

7. API Integration Catalog

The A2A system integrates with the following external APIs. All credentials are managed via Google Secret Manager and rotated quarterly.

Category	API / Service	Used By Agent	Purpose
Flights	Amadeus GDS	Flight Agent	Flight search and booking
Flights	Google Flights	Flight Agent	Price intelligence, tracking
Hotels	Booking.com API	Hotel Agent	Hotel search and booking
Hotels	Google Hotels	Hotel Agent	Location-based discovery
Transport	Uber for Business	Transport Agent	Cab bookings, receipts
Transport	Google Maps Platform	Guide + Transport	Routes, places, directions
Visa	Sherpa API	Visa Agent	Visa requirement lookup
Weather	OpenWeatherMap	Guide + Safety Agent	Forecasts, severe weather
Flights Monitor	FlightAware API	Safety Agent	Real-time flight status
Communication	Twilio	Notify Agent	SMS and voice calls
Communication	SendGrid	Notify Agent	Email delivery
OCR / Vision	Gemini Vision API	Expense Agent	Receipt OCR and parsing
Payments	Razorpay / Visa API	Expense Agent	Corporate card sync
Analytics	Google Looker Studio	Dashboard	Reports and visualizations
Insurance	Travel Insured API	Visa + Guide Agent	Policy validation, claims

8. Recommended Additions & Enhancements

The following features were identified during design as high-value additions not in the original scope. These are recommended for inclusion in subsequent releases.

Note to Operations Team

This section captures gaps and enhancements identified by the design team. Prioritize these based on operational pain points and bring them to the product roadmap discussion.

8.1 Identified Gaps in Original Scope

#	Missing Feature	Why It Matters	Suggested Agent
1	Pre-trip approval workflow	Prevents unauthorized travel; enforces manager sign-off before any booking	Orchestrator + Notify
2	Travel advance / cash float management	Travelers often need cash advances before trips; no mechanism existed	Expense Agent
3	Meeting and agenda coordination	Client visit agenda should link to travel logistics for timing accuracy	New: Calendar Agent
4	Language and translation support	International visits need on-demand translation for traveler safety	Guide Agent extension
5	Duty of care / wellness check-ins	Periodic SMS check-ins during trip; auto-escalate if no response	Safety Agent
6	CO2 emissions tracking	ESG reporting requires carbon footprint data per trip	New: Sustainability Agent
7	Multi-traveler group trips	Multiple employees traveling together should share one trip record	Orchestrator upgrade
8	Loyalty points consolidation	Corporate points across airlines and hotels are not tracked or redeemed	Flight + Hotel Agents
9	Vendor invoice reconciliation	Invoices from hotels/airlines must match booking system records	Expense Agent
10	Mobile offline mode	Travelers in poor connectivity areas need offline itinerary access	Mobile App layer

#	Missing Feature	Why It Matters	Suggested Agent
11	Spousal / dependent travel policy	Some client visits allow family extension; policy and tracking needed	Policy Engine
12	Post-visit debrief capture	No mechanism to capture trip outcome, client feedback, or follow-ups	New: Debrief Agent

8.2 Proposed Phase 2 Agents

Calendar & Meeting Coordination Agent

Synchronizes the trip itinerary with Google Calendar events for the client visit. Automatically buffers travel time between flight arrival and first meeting, sends reminders, and blocks focus time for pre-meeting preparation. Integrates with the client's shared calendar when access is granted.

Sustainability & ESG Reporting Agent

Calculates carbon emissions for every trip leg (flight distance, transport type, hotel energy rating). Generates monthly and annual ESG travel reports with per-employee footprint, department benchmarks, and offset purchase recommendations aligned with organizational sustainability targets.

Vendor Management Agent

Manages relationships with preferred hotels, airlines, and transport vendors. Tracks negotiated rate compliance, renewal deadlines for corporate agreements, and vendor performance scores based on booking outcomes. Flags underperforming vendors and recommends replacements.

Post-Visit Debrief Agent

Within 48 hours of trip return, prompts the traveler with a structured debrief covering: meeting outcomes, client sentiment score, follow-up actions, relationship status update, and trip ROI notes. Feeds data to the CRM and provides the sales team with ready-made client visit summaries.

9. Deployment Guide

9.1 Prerequisites

- Google Cloud Project with billing enabled
- Vertex AI API, Cloud Run, Firestore, BigQuery, Pub/Sub, Secret Manager APIs enabled
- Domain-verified Google Workspace account for IAP and OAuth
- Service accounts created for each agent with minimum required IAM permissions
- All external API keys (Amadeus, Booking.com, Twilio, etc.) provisioned and stored in Secret Manager

9.2 Deployment Sequence

22. Infrastructure Setup: Create VPC, subnets, Cloud Run service accounts, and Firestore database in chosen region
23. Agent Deployment: Deploy each agent container to Cloud Run; configure min/max instances per traffic profile
24. Agent Registry: Register all Agent Cards with Vertex AI Agent Builder; test discovery endpoints
25. Pub/Sub Topics: Create topics for each agent communication channel; configure dead-letter queues
26. Policy Configuration: Upload corporate travel policy rules to Firestore policy collection
27. Dashboard Setup: Connect BigQuery datasets to Looker Studio; configure operations dashboards
28. Mobile App: Deploy traveler mobile app (Flutter) with Firebase Auth; configure push notification channels
29. UAT: Execute end-to-end test trips covering all agent interactions, expense flows, and exception scenarios
30. Operations Onboarding: Train operations team on dashboard, approval workflows, and exception handling
31. Phased Rollout: Start with a pilot team of 10 travelers; expand to full organization post-validation

9.3 Estimated Timeline

Phase	Duration	Deliverables
Phase 0 — Foundation	2 weeks	GCP infrastructure, IAM, VPC, Secret Manager, CI/CD pipelines
Phase 1 — Core Agents	6 weeks	Flight, Hotel, Transport, Expense Agents; basic orchestration

Phase	Duration	Deliverables
Phase 2 — Intelligence	4 weeks	Guide, Safety, Visa, Notification Agents; full A2A routing
Phase 3 — Operations	3 weeks	Dashboard, approval workflows, mobile app, policy engine
Phase 4 — UAT & Pilot	3 weeks	End-to-end testing, operations onboarding, pilot with 10 users
Phase 5 — Full Rollout	2 weeks	Organization-wide deployment, analytics, Phase 2 agent planning
TOTAL	~20 weeks	Production-ready A2A Travel Management System

10. Glossary

Term	Definition
A2A Protocol	Agent-to-Agent — Google's open protocol for multi-agent AI system communication
Agent Card	JSON manifest (/.well-known/agent.json) describing an agent's capabilities, inputs, and endpoints
GDS	Global Distribution System — airline reservation network (e.g., Amadeus, Sabre)
Gemini	Google's family of large language models powering each agent's reasoning
IAP	Identity-Aware Proxy — Google's zero-trust access control for Cloud Run services
MCC	Merchant Category Code — 4-digit code classifying a merchant's business type for payment categorization
Orchestrator	The master AI agent that receives all user inputs and delegates tasks to specialist agents
Per Diem	Daily allowance for meals and incidentals, set by company policy per destination tier
Pub/Sub	Google Cloud Pub/Sub — managed message queue for async agent-to-agent task passing
SSE	Server-Sent Events — streaming protocol used by A2A for real-time task progress updates
Task Object	Standardized A2A data structure containing task ID, session, message parts, and metadata
Vertex AI	Google Cloud's unified AI/ML platform hosting agents, models, and the agent registry

END OF DOCUMENT

A2A Travel Management System | Technical Documentation v1.0 | Internal Use Only